Music Sound Interpreter

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The Rundown

- Who is the client?
- What is the project?
- Why is this being done?
- How will this project be done?
- How much time will it take?
- How much is this project going to cost?

The Client

- My client is Rosemary
  - Has Down Syndrome
  - Is deaf
  - Communicates by sign language
  - Spends some of her time at Bridgewell in Lynn

Some Facts about Down Syndrome

- Down Syndrome is a genetic disorder involving the 21st chromosome
- About 1 in 700 people are born with DS
- 400,000 diagnosed people in the U.S.
- Most prominent type is Trisomy 21
  - Accounts for 95% of all cases
  - Cells have an extra copy of chromosome 21

Effects of Down Syndrome

- Effects vary from person to person
- Possible for diagnosed people to lead normal lives
- Common effects include:
  - Physical disfigurement
  - Slow mental development
  - Increased risk of health issues

The Problem

- Bridgewell hosts learning activities such as concerts
- Since Rosemary is deaf, she feels the speakers to understand what's going on
- In doing so, she is physically left out of her groups

The Solution: My Project

- Lap pad that senses audio signals then activates visual effects on one side and vibrational effects on the other
- Contains recorded responses such as: ”That's awesome!”
- Using the Music Sound Interpreter, she will able to join her groups extensively

Functions

- Visual and vibrational effects controlled by frequency and strength of audio signals
- Visual components consist of a rectangular LED grid
  - Each column represents a different frequency range
  - As intensity changes, LEDs fade in and out
- Vibrational effects activated similarly

Bonus Purposes

- Can be used to teach individuals how to control their voices
  - Instructor would say something to be copied,
    then the user would then try to match the instructor
- Could also be used as decorative piece

Designing for Everyone

- Rosemary will most likely not be the only user
- Should be designed so it's usable by anyone
- Design factors include:
  - Safety
  - Accessibility
  - Simplicity

Physical Specifications

- About as big as an open notebook (12”x21”x1”)
- Weight = 5 lbs
  - The lighter, the better
- Cushioned
- Shatterproof
- Waterproof
- Easily cleanable
- Cased in flexible material

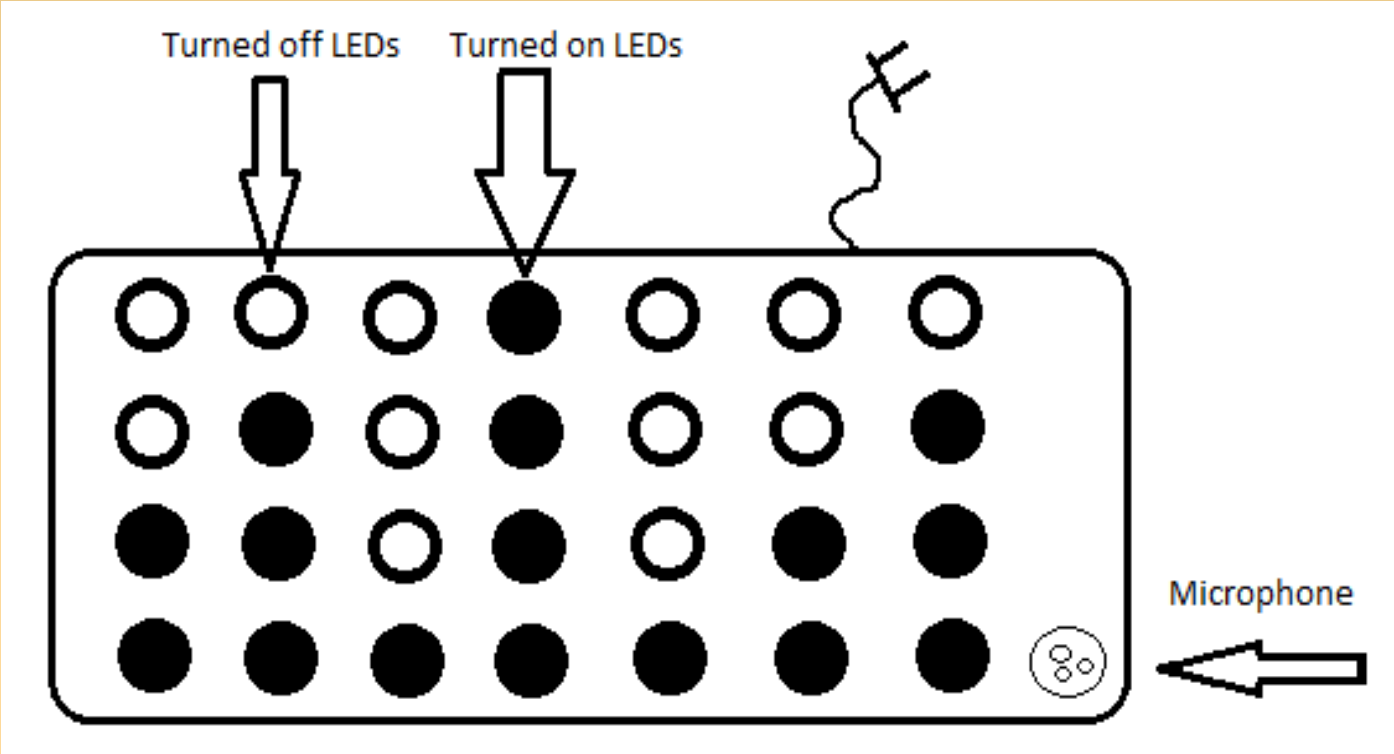
Electrical Specifications

- Powered via wall outlet
- Includes a function toggle switch
  - Only vibrations, only lights, both on, both off
- Automatic shut-off feature
- Low voltage, about 12V maximum
- Medium current, about 2A maximum

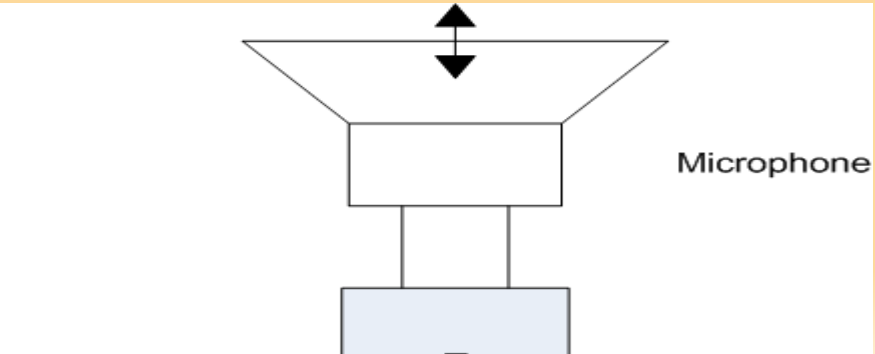
Effect Specs

- Uses calming colors such as pink, purple and green
  - Avoid using red, yellow, and orange
- Uses smooth visual components
  - No fast-blinking lights
- Intensity of lights and vibrations shouldn't be strong

Functional Diagram



Block Diagram

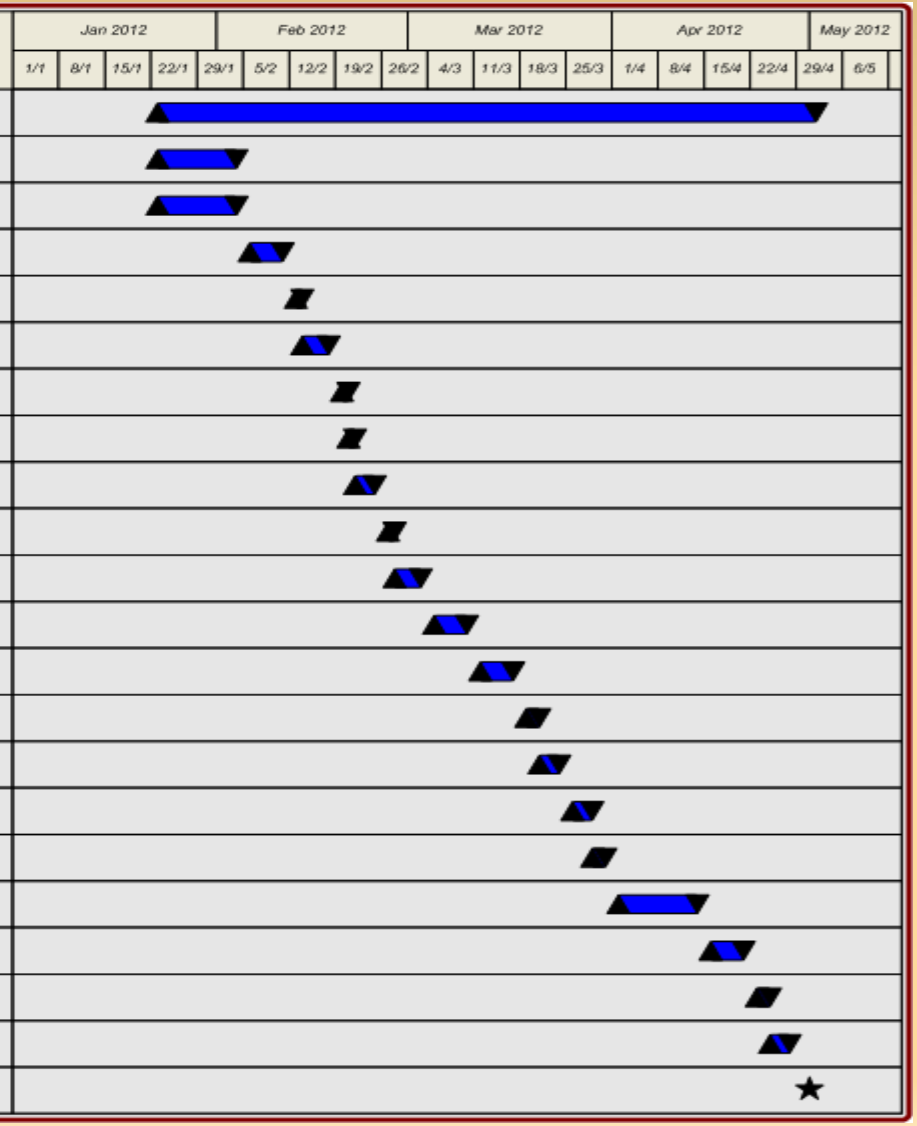


Creation Procedure

- Design parts from beginning of product activation towards output
- Test parts in descending order
  - First tests are stand alone
  - Second are with previous parts included

Gantt Chart

ID	Task Name	Start	Finish	Duration	Hours Worked
1	Total Completion Time	1/23/2012	5/1/2012	100d	193h
2	Part Hunting	1/23/2012	2/3/2012	12d	20h
3	Design General Final Circuit	1/23/2012	2/3/2012	12d	20h
4	Pre-Amp Design	2/6/2012	2/10/2012	5d	10h
5	Test Pre-Amp connected to Mic	2/13/2012	2/13/2012	1d	4h
6	Filter Design	2/14/2012	2/17/2012	4d	8h
7	Stand Alone Filter Test	2/20/2012	2/20/2012	1d	4h
8	Filter test with Pre-amp and Mic	2/21/2012	2/21/2012	1d	5h
9	LED Controller Design	2/22/2012	2/24/2012	3d	16h
10	Stand Alone LED Controller Test	2/27/2012	2/27/2012	1d	4h
11	Motor Controller Design	2/28/2012	3/2/2012	4d	8h
12	Stand Alone Motor Controller Test	3/5/2012	3/9/2012	5d	10h
13	Main Circuit Test	3/12/2012	3/16/2012	5d	10h
14	Design and Test Voice Module	3/19/2012	3/20/2012	2d	10h
15	Design Automatic Shut-Off Feature	3/21/2012	3/23/2012	3d	12h
16	Stand Alone Shut-Off Test	3/26/2012	3/28/2012	3d	6h
17	Shut-Off Test with Main Circuit	3/29/2012	3/30/2012	2d	4h
18	Design Supply Circuit	4/2/2012	4/13/2012	12d	20h
19	Test Final Circuit	4/16/2012	4/20/2012	5d	10h
20	Assemble Final Product	4/23/2012	4/24/2012	2d	4h
21	Test Final Product	4/25/2012	4/27/2012	3d	6h
22	Present Music Sound Interpreter	5/1/2012	5/1/2012	0d	



Time Analysis

- Beginning Date: January 23, 2012
- Ending Date: May 1, 2012
- Duration: 100 days
- Time spent on project: 193 hours
- Designing each block mostly takes up a week
- Testing usually takes a few days
- Work between 2-4 hours per day

Market Research

- Most products are very expensive
- Not many found that have similar functions
- Most products have simple, “discrete” functions
- Very few are sound controlled
- Potential for large market, even outside of assistive technology

Main Competing Product

Interactive Fanlight

- Displays lights in a fan-like manner
- Automatic and sound controlled modes
- Multiple display styles within each mode
- Only amplitude dependent
- Priced at $2,675.00

Labor and Overhead

- Average time worked per week: 13.5 hours
- 14 week long project
- Labor Cost = $5,724
- Overhead Cost= $2,919

Parts and Prices

Total Cost

- Direct Labor: $5,724
- Overhead: $2,919
- Parts: $325
- **Total Cost of Project: $8,962**

Final Words

Any Questions?

Thanks everyone for your time!